# 10.4: The Calvin cycle uses the chemical energy of ATP and NADPH to reduce $CO_2$ to sugar

The Calvin cycle takes place in the stroma; it is similar to the citric acid cycle in that a starting material is regenerated after some molecules enter and others exit the cycle. However, the citric acid cycle is catabolic, oxidizing acetyl CoA and using the energy to synthesize ATP (see Figure 9.11), while the Calvin cycle is anabolic, building carbohydrates from smaller molecules and consuming energy. Carbon enters the Calvin cycle in the form of $CO_2$ and leaves in the form of sugar. The cycle spends ATP as an energy source and consumes NADPH as reducing power for adding high-energy electrons to make the sugar.

As we mentioned in Concept 10.2, the carbohydrate produced directly from the Calvin cycle is not glucose. It is actually a three-carbon sugar called **glyceraldehyde 3-phosphate(G3P)**. For the net synthesis of *one* molecule of G3P, the cycle must take place three times, fixing *three* molecules of $CO_2$—one per turn of the cycle. (Recall that the term *carbon fixation* refers to the initial incorporation of $CO_2$ into organic material.) As we trace the steps of the Calvin cycle, keep in mind that we are following three molecules of $CO_2$ through the reactions. **Figure 10.18** divides the Calvin cycle into three phases: carbon fixation, reduction, and regeneration of the $CO_2$ acceptor.

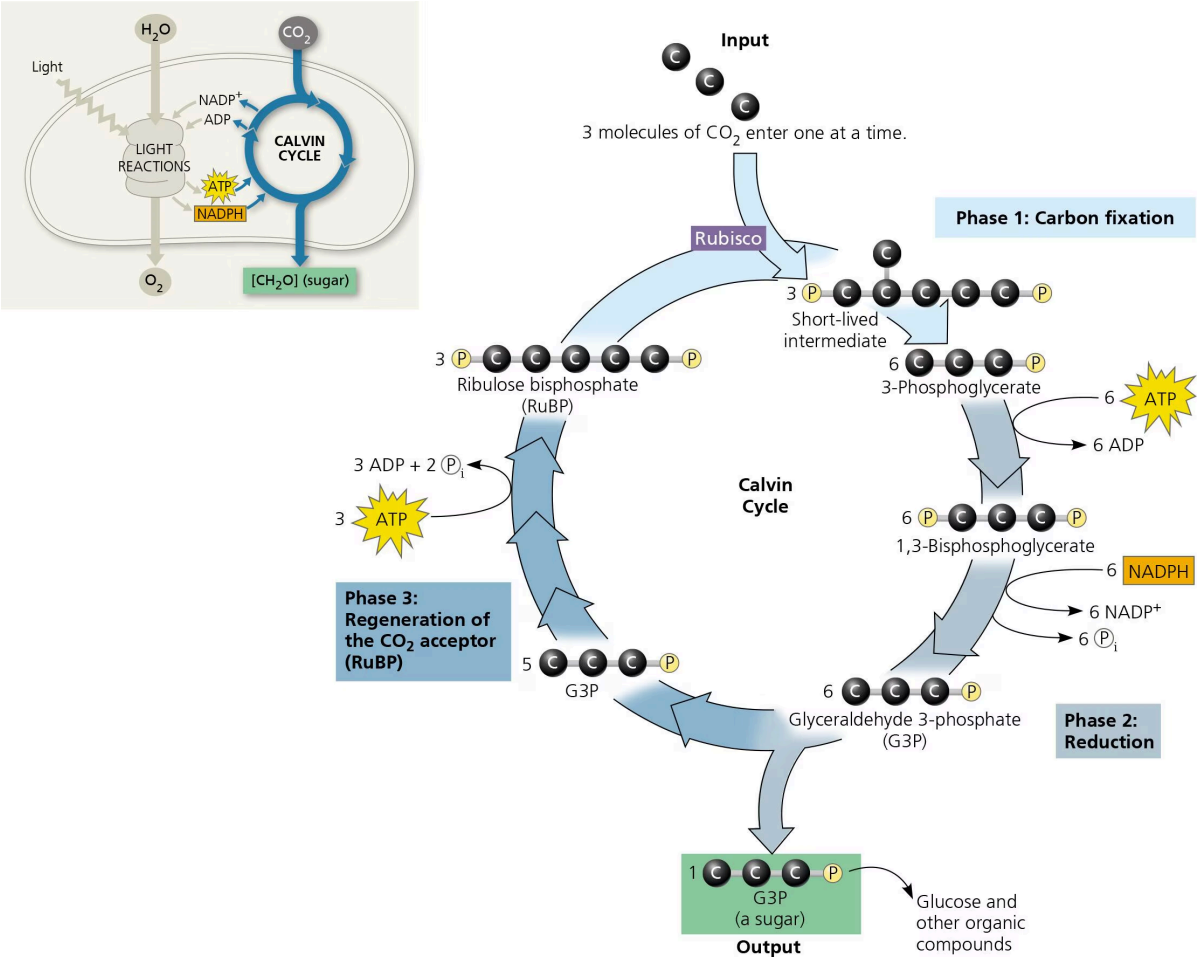
- - ******Phase 1:****Carbon fixation**. The Calvin cycle incorporates each $CO_2$ molecule, one at a time, by attaching it to a five-carbon sugar named ribulose bisphosphate (which is abbreviated RuBP). The enzyme that catalyzes this first step is RuBP carboxylase-oxygenase, or **rubisco** (see Figure 6.32c). (This is the most abundant protein in chloroplasts and is also thought to be the most abundant protein on Earth.) The product of the reaction is a six-carbon intermediate that is short-lived because it is so energetically unstable that it immediately splits in half, forming two molecules of 3-phosphoglycerate (for each $CO_2$ fixed).
- - ******Phase 2:****Reduction**. Each molecule of 3-phosphoglycerate receives an additional phosphate group from ATP, becoming 1, 3-phosphoglycerate. Next, a pair of electrons donated from NADPH reduces 1, 3-phosphoglycerate, which also loses a phosphate group in the process, becoming glyceraldehyde 3-phosphate (G3P). Specifically, the electrons from NADPH reduce a carboxyl group on 1, 3-bisphosphoglycerate to the aldehyde group of G3P, which stores more potential energy. G3P is a sugar—the same three-carbon sugar formed in glycolysis by the splitting of glucose (see Figure 9.8). Notice in Figure 10.18 that for every *three* molecules of $CO_2$ that

enter the cycle, there are *six* molecules of G3P formed. But only one molecule of this three-carbon sugar can be counted as a net gain of carbohydrate because the rest are required to complete the cycle. The cycle began with 15 carbons' worth of carbohydrate in the form of three molecules of the five-carbon sugar RuBP. Now there are 18 carbons' worth of carbohydrate in the form of six molecules of G3P. One molecule exits the cycle to be used by the plant cell, but the other five molecules must be recycled to regenerate the three molecules of RuBP.

- - ****Phase 3:**Regeneration of the*$CO_2$**acceptor (RuBP).** In a complex series of reactions, the carbon skeletons of five molecules of G3P are rearranged by the last steps of the Calvin cycle into three molecules of RuBP. To accomplish this, the cycle spends three more molecules of ATP. The RuBP is now prepared to receive $CO_2$ again, and the cycle continues.

**Figure 10.18**

The Calvin cycle.

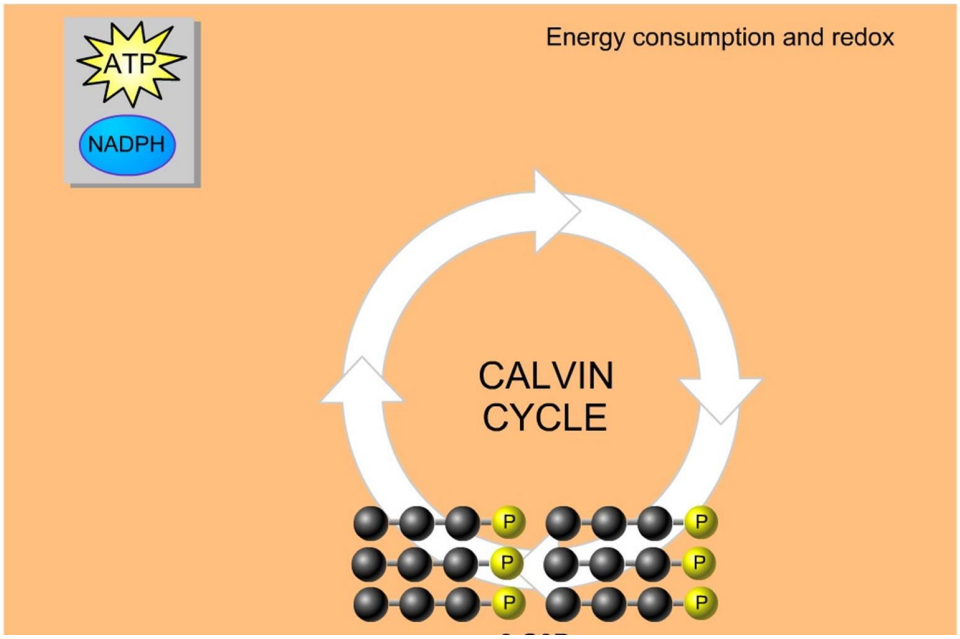


Input. 3 molecules of C O 2 enter one at a time. Phase 1, carbon fixation. Rubisco becomes a short-lived carbon intermediate with phosphates. It then becomes 6 3-phosphoglycerate. 6 A T P enter, and 6 A D P exit. 6 1, 3-biphosphoglycerate is formed. 6 N A D P H enter and 6 N A D P plus exit, plus another 6 P subscript i exit. Phase 2, reduction. 6 glyceraldehyde 3-phosphate is formed. They have an output of G 3 P, a sugar, with glucose and other organic compounds. Within the cycle, 5 G 3 P continues. Phase 3, regeneration of the C O 2 acceptor (RuBP). 3 A T P enter, 3 A D P plus 2 P subscript i exit. 3 ribulose biphosphate (RuBP) are formed. These continue to where the original input of 3 molecules of C O 2 form rubisco.

This diagram summarizes three turns of the cycle, tracking carbon atoms (shown as dark-gray spheres). The three phases of the cycle correspond to the phases discussed in the text. For every three molecules of $CO_2$ that enter the cycle, the net output is one molecule of glyceraldehyde 3-phosphate (G3P), a three-carbon sugar. The light reactions sustain the Calvin cycle by regenerating the required ATP and NADPH.

**Animation**

The Calvin Cycle

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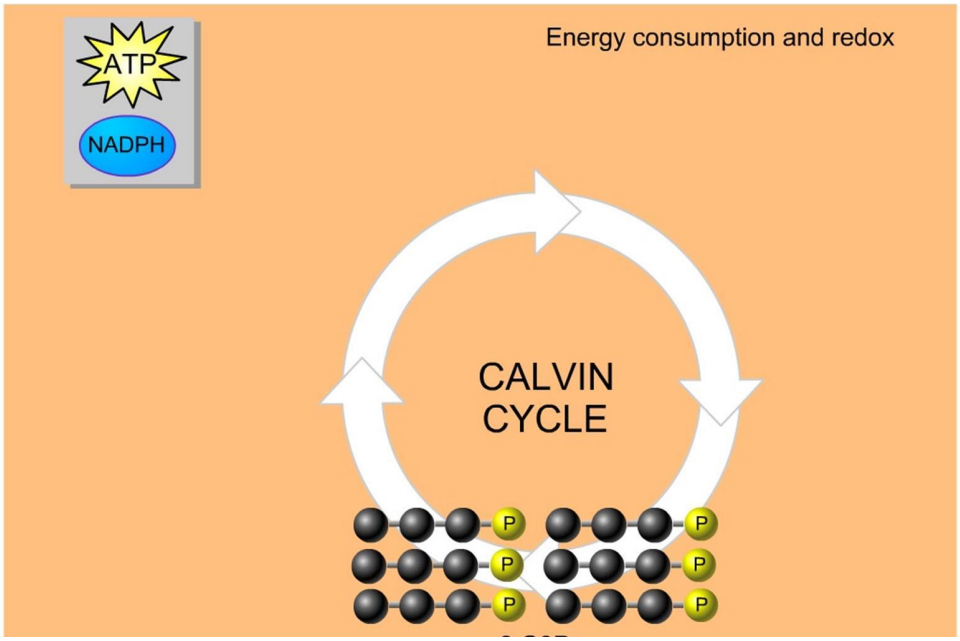


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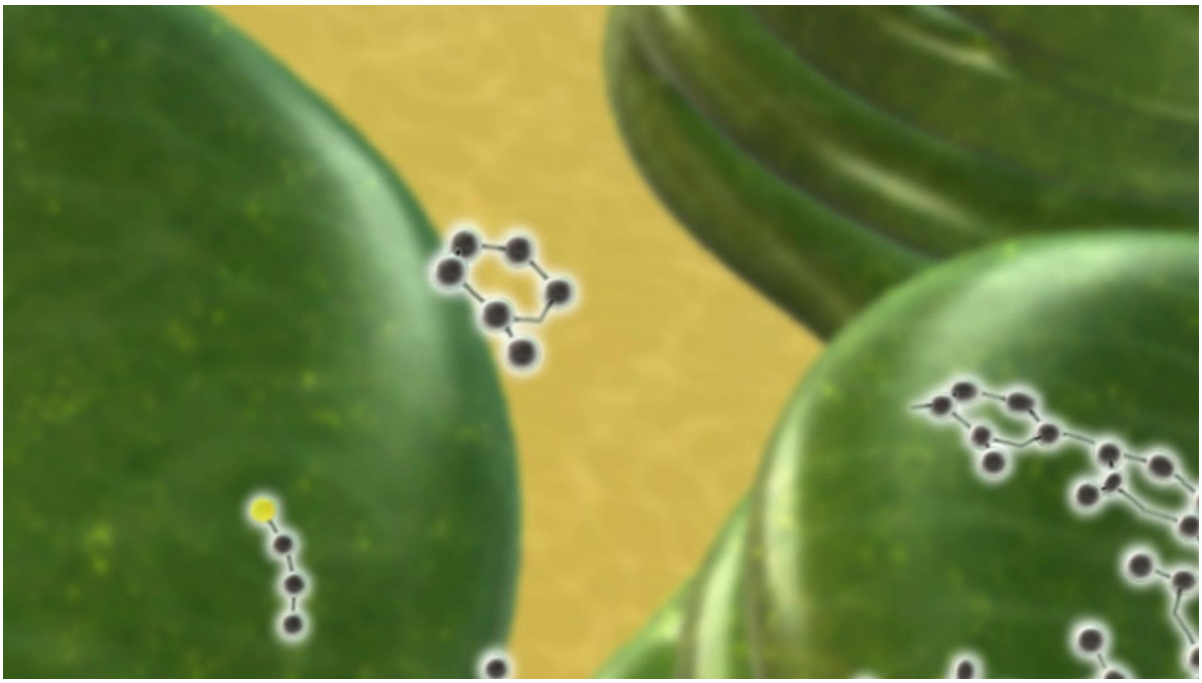
The Calvin Cycle



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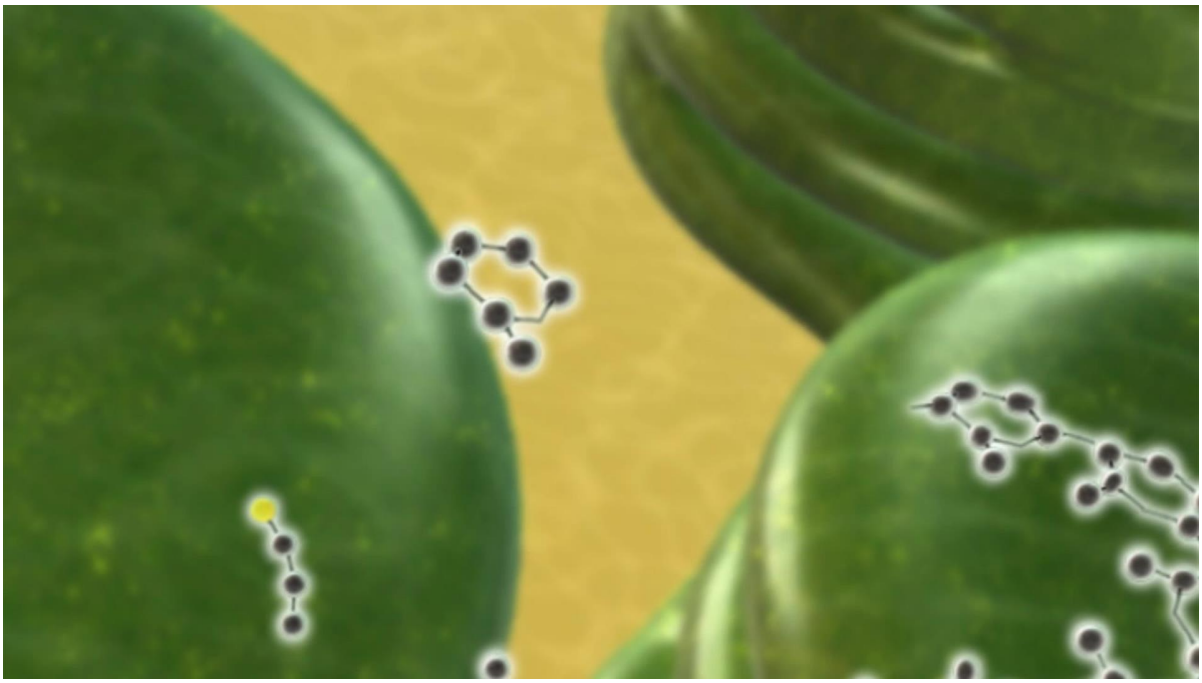

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For the net synthesis of one G3P molecule, the Calvin cycle consumes a total of nine molecules of ATP and six molecules of NADPH. The light reactions regenerate the ATP and NADPH. The G3P spun off from the Calvin cycle becomes the starting material for metabolic pathways that synthesize other organic compounds, including glucose (from two molecules of G3P), sucrose, and other carbohydrates. Neither the light reactions nor the Calvin cycle alone can make sugar from $CO_2$. Photosynthesis is an emergent property of the intact chloroplast, which integrates the two stages of photosynthesis.